Chapter 13

Error Analysis on Triangles

13.1 Outline

We shall carry out a quantitative error analysis of the Ritz-Galerkin method. The analysis will be valid for a certain class of problems. First of all, we assume that the assumptions of the Lax-Milgram theorem are satisfied. Furthermore, for simplicity and definiteness we focus on

- 2D problems, posed in bounded polygonal domains Ω;
- problems posed in $H_0^1(\Omega)$;
- an approximation space S_h consisting of piecewise linear approximation on triangular grids.

Most of these assumptions pose no essential restriction and they are made mainly for definiteness. For example, the analysis can be done in three spatial dimensions using tetrahedral grids in a very similar manner. Focusing on problems posed in $H_0^1(\Omega)$ makes sense for second order elliptic problems with homogeneous boundary conditions. These are the kind of problems we have discussed so far. Extensions will be discussed later. Restricting ourselves to polygonal domains Ω seems odd, but we shall show in a later chapter that for piecewise linear approximations the error committed by approximating a general domain by a polygonal boundary (i.e. by the external boundary of a triangulation) does not introduce an error larger than the approximation error. (For higher-degree polynomial approximations this is no longer true, and one needs to be more carful. However, we do not discuss this here.)

Assume that we are able to define a characteristic length scale h that we can use to measure the level of fineness of each mesh that we use. (We discuss below how to do this.) Our aim is to estimate how fast the error will decrease as a power of h. For example, we prove for the error of the Ritz-Galerkin approximation an inequality of the type

$$\|u - u_h\|_{0,\Omega} = \mathcal{O}(h^\alpha), \tag{13.1}$$

where C, α are positive constants. However, it will be more straightforward to first produce an expression for the error using the norm $\|\cdot\|_{1,\Omega}$. The reason is that we can establish for this norm a chain of inequalities starting from the estimates that we already have:

$$\|u - u_h\|_{1,\Omega} \le c_1 \inf_{v_h \in S_h} \|u - v_h\|_{1,\Omega} \le c_1 \|u - \mathcal{I}_h u\|_{1,\Omega} = \mathcal{O}(h^\beta), \tag{13.2}$$

where S_h is our approximation space and $\mathcal{I}_h$ is an interpolation operator into S_h, to be defined below. The first inequality is already established (with $c_1 = C/\alpha$) by Theorem 12.2. This inequality holds for any approximation of the Ritz-Galerkin type, provided that the assumptions of Theorem 12.2 are satisfied. (Note carefully that there is no such result for the L^2-norm $\|\cdot\|_{0,\Omega}$!) Naturally, we can bound this optimal approximation by the interpolation error $\|u - \mathcal{I}_h u\|_{1,\Omega}$. (The interpolation error cannot be smaller than the *best* approximation in S_h.) This interpolation error is *independent* of the PDE problem under consideration, and depends only on the

approximation space[1]. This is the second inequality. Showing the last inequality in (13.2) is the goal of this chapter. We shall see that $\beta = 1$ for piecewise linear approximation. After we have established that bound, we show that under mild assumptions the error in (13.1) is of second order, i.e. $\alpha = 2$.

13.2 Triangulation

We define a triangle as the closed set (i.e including the boundary) given by the convex hull of exactly three points, which we call *vertices*. The straight line connecting two vertices is called an *edge*. (More general definitions can be given, which allow triangles to have curved edges.) A triangulation, sometimes called a *mesh*, is a decomposition of the domain Ω into triangles. We first give a very basic definition of an *admissible* triangulation:

Definition 13.1 *We call a triangulation $\mathcal{T}_h = \{T_1, \ldots, T_N\}$ of a domain Ω admissible if*

1. $\overline{\Omega} = \bigcup_{i=1}^{N} T_i$.
2. *If, for any $i \neq k$, the intersection $T_i \cap T_k$ is not empty, that intersection is either an edge of both triangles or a vertex of both triangles.*

Going back to Fig. 9.1, for example, we would see that for the purposes of this section only the configuration a) would be admissible. The reason for this choice of rules will become apparent below.

For the convergence analysis below, where we deal with families of triangulations, rather than just one specific triangulation[2], we shall need one more rule. For now, however, this is sufficient to go on.

13.3 The Interpolation Operator

Consider a problem posed in H^1 with homogeneous boundary conditions, such as the weak form of Model Problem 4. Let $\mathcal{T}_h$ be an admissible triangulation, and define

$$S_h := \{v \in C^0(\Omega) \,:\, v|_T \in P^1(T)\ \forall T \in \mathcal{T}_h;\ v(x) = 0,\ x \in \partial\Omega\} \tag{13.3}$$

Here $P^1(T)$ is the space of all affine linear polynomials defined on T. The approximation space thus consists of such functions that are continuous in all of Ω, and whose restriction to each triangle is affine linear. Note that $S_h \subset H_0^1(\Omega)$, as required by the Ritz-Galerkin method. It is worth pointing out that we have assumed that the boundary of Ω is polygonal and we can thus assume that the boundary of the triangulation coincides with that of Ω. This makes it easy to satisfy the boundary conditions and thus ensure $S_h \subset H_0^1(\Omega)$. In a later chapter we will investigate the consequences of violating this constraint. We define an *Interpolation operator* first for a single triangle T. That interpolation operator is a mapping $\mathcal{I}_T : C(T) \to P^1(T)$. Let $v \in C(T)$, and z_k be the vertices of the triangle T, where k is in some appropriate index set I_T established by the global numbering for the vertices of the mesh. Specifying the values at the three vertices uniquely specifies the interpolant. The interpolation operator is defined as

$$(\mathcal{I}_T v)(x) := \sum_{k \in I_T} v(z_k)\phi_k(x) \tag{13.4}$$

[1]There is an implicit connection to the solution of the PDE problem, because the interpolation error depends on the smoothness of the function to be approximated. The quantitative results can thus be used only if the solution of the PDE problem is smooth enough.

[2]Recall that for the convergence analysis we consider sequences of approximations on successively finer grids.

where the $\phi_k \in P^1(T)$ are basis functions that satisfy $\phi_k(z_i) = \delta_{ik}$, where k ranges over all interior vertices, and i over all vertices. (Note that for vertices on the boundary $v(z_k) = 0$, and no basis functions for such nodes need to be defined. The index set I_T thus includes only interior vertices.) The global interpolation operator $\mathcal{I}_h$ is then uniquely determined by simply requiring

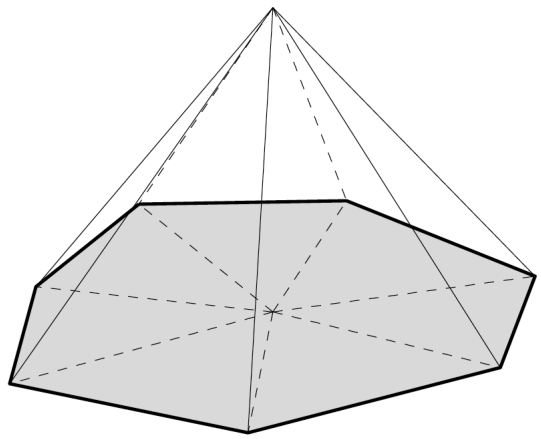

Figure 13.1: Graphical illustration of basis functions for the approximation space (13.3) ("hat functions"). Each basis function is associated with a vertex z_k. Outside the patch of triangles shown here, the basis function is identically zero

that the restriction to each triangle be given by (13.4), i.e. $\mathcal{I}_h v|_T = \mathcal{I}_T v$ for all $T \in \mathcal{T}_h$. It turns out that it is enough to define such a local operator for only one triangle, termed the reference triangle, and get an interpolation operator for any triangle in the mesh by using an appropriate mapping.

13.4 The Reference Element

We define a *reference triangle* as[3]

$$\hat{T} := \{(\xi, \eta) : \xi, \eta \geq 0,\ \xi + \eta \leq 1\}$$

We denote coordinates $\hat{x} := (\xi, \eta)$. Note that the vertices of the reference triangle are $\hat{z}_1 = (0, 0)$, $\hat{z}_2 = (1, 0)$, and $\hat{z}_3 = (0, 1)$. Let $\hat{v}$ be a function defined on $\hat{T}$. The interpolation operator (13.4) for this triangle can be written

$$(\mathcal{I}_T \hat{v})(\hat{x}) := \sum_{i=1}^{3} v(\hat{z}_i)\hat{\phi}_k \tag{13.5}$$

where

$$\begin{aligned} \hat{\phi}_1 &= 1 - \xi - \eta \\ \hat{\phi}_2 &= \xi \\ \hat{\phi}_3 &= \eta. \end{aligned}$$

Convince yourself that the basis functions $\hat{\phi}_k$ satisfy $\phi_k(z_i) = \delta_{ki}$ for $i, k = 1, 2, 3$. We define a mapping

$$F_T : \hat{T} \to T,$$

[3]The choice of reference triangle is not unique. But once chosen, it will remain fixed throughout the analysis.

which maps the points in the reference triangle to a triangle $T \in \mathcal{T}_h$, via affine function

$$T \ni (x, y) = F_T(\hat{x}) = b + B\hat{x}. \tag{13.6}$$

The Jacobian B is thus composed of all the partials

$$B = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{pmatrix}$$

An explicit form of such a mapping, which maps each $(\xi, \eta) \in \hat{T}$ to $(x, y) \in T$ can be written, denoting again the vertices of the triangle T as z_i, for $i = 1, 2, 3$ (we use local indices for convenience):

$$F_T = z_1 + (z_2 - z_1)\xi + (z_3 - z_1)\eta.$$

Comparing to the generic form (13.6), and introducing the x and y components of the vertices $z_i = (z_i^{(x)}, z_i^{(y)})$, we find that

$$b = \begin{pmatrix} z_1^{(x)} \\ z_1^{(y)} \end{pmatrix}, \qquad B = \begin{pmatrix} z_2^{(x)} - z_1^{(x)} & z_3^{(x)} - z_1^{(x)} \\ z_2^{(y)} - z_1^{(y)} & z_3^{(y)} - z_1^{(y)} \end{pmatrix}.$$

Note that the determinant $\det B$ is just the cross product of the two edges. Hence the magnitude of the determinant is twice the volume of the triangle: $|\det B| = 2|T|$. This implies that the transformation is invertible for any triangle with non-zero volume, and we can define (ξ, η) as a function of (x, y) via the inverse mapping

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = B^{-1} \left\{ \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} z_1^{(x)} \\ z_1^{(y)} \end{pmatrix} \right\},$$

where the inverse of B is given by

$$B^{-1} = \frac{1}{\det B} \begin{pmatrix} z_3^{(y)} - z_1^{(y)} & -(z_3^{(x)} - z_1^{(x)}) \\ -(z_2^{(y)} - z_1^{(y)}) & z_2^{(x)} - z_1^{(x)} \end{pmatrix}.$$

The interpolation for the element T is thus given by (13.4) with the restriction of the basis functions to T defined as

$$\begin{aligned}
\phi_1(x, y) &= 1 - \xi(x, y) - \eta(x, y) \\
&= 1 - \frac{1}{\det B} \left\{ (z_3^{(y)} - z_1^{(y)})(x - z_1^{(x)}) - (z_3^{(x)} - z_1^{(x)})(y - z_1^{(y)}) \right\} \\
&\quad + \frac{1}{\det B} \left\{ (z_2^{(y)} - z_1^{(y)})(x - z_1^{(x)}) - (z_2^{(x)} - z_1^{(x)})(y - z_1^{(y)}) \right\} \\
\phi_2(x, y) &= \xi(x, y) \\
&= \frac{1}{\det B} \left\{ (z_3^{(y)} - z_1^{(y)})(x - z_1^{(x)}) - (z_3^{(x)} - z_1^{(x)})(y - z_1^{(y)}) \right\} \\
\phi_3(x, y) &= \eta(x, y) \\
&= \frac{-1}{\det B} \left\{ (z_2^{(y)} - z_1^{(y)})(x - z_1^{(x)}) - (z_2^{(x)} - z_1^{(x)})(y - z_1^{(y)}) \right\}
\end{aligned}$$

Convince yourself that the property $\phi_i(z_j) = \delta_{ij}$ holds. Note that, although we have not explicitly indicated this in our notation, the vertex coordinates and the Jacobian of the mapping are different for each triangle. These quantities, however, can be easily computed from a given triangulation using the data structure normally at hand (volumes, vertex coordinates.)

In the following, interpolation errors are also computed on the reference element, and the error on the triangulation is measured via the mapping we have defined. Consequently we need transformation rules for integration and differentiation (keep in mind that the Sobolev norms

require differentiation.) According to standard rule of integration, the integral over a triangle transforms as

$$\int_T dxdy = \int_{\hat{T}} |\det B| d\xi d\eta = |\det B| \int_{\hat{T}} d\xi d\eta$$

where we have used the fact that the Jacobian of the mapping is independent of (x, y). Hence

$$\|u\|_{0,T} = \sqrt{|\det B|}\, \|\hat{u}\|_{0,\hat{T}}$$

For higher-order Sobolev norms, we also need to measure the L^2 norm of *derivatives*. First note that, by chain rule, the partials transform

$$\frac{\partial \hat{u}}{\partial \xi} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \xi} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \xi}$$
$$\frac{\partial \hat{u}}{\partial \eta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \eta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \eta}$$

and so, using the Jacobian of the mappring (13.6), the gradient transforms as

$$\nabla_{\hat{x}} \hat{u} = B^T \nabla_x u,$$

where B^T denotes the transpose of the Jacobian. So the H^1 semi-norm transforms as

$$|\hat{u}|^2_{1,\hat{T}} = \int_{\hat{T}} |\nabla_{\hat{x}} \hat{u}|^2 d\xi d\eta = \int_T |B^T \nabla_x u|^2 \frac{dxdy}{|detB|}.$$

Using standard matrix norm inequality[4], $|B^T \nabla_x u| \leq \|B\|\, |\nabla_x u|$ there follows easily

$$|\hat{u}|_{1,\hat{T}} \leq \|B\| \frac{1}{\sqrt{\det B}} |u|_{1,T}$$

Higher order norms follow similar transformations. Without proof we state:

Theorem 13.1 *Let $\hat{T}$ and T be polygons related by the affine transformation* (13.6)*. Then there exist constants $\hat{C}, C$ such that*

$$|u|_{m,T} \leq \hat{C} \left\|B^{-1}\right\|^m \sqrt{|\det B|} |\hat{u}|_{m,\hat{T}} \tag{13.7}$$

$$|\hat{u}|_{m,\hat{T}} \leq C \|B\|^m \frac{1}{\sqrt{|\det B|}} |u|_{m,T} \tag{13.8}$$

13.5 Families of triangulations

In order to give convergence estimates, we need not only one triangulation, but rather a sequence of triangulations with more and more, and increasingly smaller, elements. Convergence analysis produces a statement as to how fast the error decreases with the "size" of the elements. This refinement cannot be done in a completely arbitrary manner. We first need a reasonable characterization of the size of the elements, meaning a length scale h that applies to each triangulation of the family. We establish simple criteria to determine whether a family of triangulations is a reasonable choice that allows us to determine how the error behaves as the mesh becomes finer.

[4]Note that for the matrix norm induced by the standard Euclidean norm $\|B\| = \left\|B^T\right\|$.

Definition 13.2 *We say a family of triangulations $\{\mathcal{T}_h\}$ is quasi-uniform, or shape-regular, if there exists a constant $\kappa > 0$, such that*

$$\frac{h_T}{\rho_T} \leq \kappa \qquad \forall T \in \mathcal{T}_h,\ \mathcal{T}_h \in \{\mathcal{T}_h\} \tag{13.9}$$

where ρ_T and h_T are the radius of the incircle, and the diameter of the triangle T, respectively. (Recall that any triangle admits a uniquely defined incircle, i.e. a circle that is tangent to all three edges. The diameter is defined as $h_T := \max_{x_1,x_2\in T} |x_1 - x_2|$.)

The significance of this definition is that the volume of a triangle going to zero implies that the diameter is going to zero as well. We cannot have "degenerate" elements, having zero volume, but non-zero edge length. This also means that for a shape-regular triangulation, the volume of every triangle can be characterized by the diameter, meaning that there exist positive constants C and C', such that

$$Ch_T^2 \leq |T| \leq C' h_T^2.$$

The second inequality is trivial, while the first follows from the fact that the area of the triangle is bound from below by the area of the in-circle, and noting that (13.9) is equivalent to

$$\rho_T \geq \frac{h_T}{\kappa}.$$

We may define an overall length scale of a triangulation by setting

$$h = diam(\Omega)^{-1} \max_{T\in\mathcal{T}_h} h_T. \tag{13.10}$$

(Scaling h with the diameter of the domain Ω ensures that $0 < h \leq 1$. This will be convenient later.) Before proceeding to the error estimates, we need a bound on the Jacobian of the mapping from the reference triangle to any triangle $T \in \mathcal{T}_h$:

Theorem 13.2 *Let $F_T : \hat{T} \to T$ be the affine map between two triangles $\hat{T}$ and T, as defined in* (13.6)*. Then one has*

$$\|B\| \leq \frac{h_T}{\rho_{\hat{T}}}, \qquad \left\|B^{-1}\right\| \leq \frac{h_{\hat{T}}}{\rho_T}. \tag{13.11}$$

Here $\|\cdot\|$ is the operator norm induced by the standard Euclidean vector norm.

Proof We prove the first inequality. The second is obtained in a similar way by considering the inverse mapping $F_T^{-1} : T \to \hat{T}$.

First note that, by definition of the norm $\|B\|$, for any $\rho > 0$, there holds

$$\|B\| = \sup_{\xi\in\mathbb{R}^2} \frac{|B\xi|}{|\xi|} = \sup_{\xi\in\mathbb{R}^2} \frac{|B\xi|}{|\xi|} \frac{\frac{\rho}{|\xi|}}{\frac{\rho}{|\xi|}} = \frac{1}{\rho} \sup_{|\eta|=\rho} |B\eta|. \tag{13.12}$$

Let $\hat{x}_a, \hat{x}_b$ be two points such that $|\hat{x}_a - \hat{x}_b| = \hat{\rho}$, but otherwise arbitrary. Now consider the image of these two points under the linear mapping F_T:

$$x_a - x_b = b + B\hat{x}_a - b - B\hat{x}_b = B(\hat{x}_a - \hat{x}_b).$$

We can assume that $\hat{x}_a, \hat{x}_b \in \hat{T}$. (Any constant offset will cancel in $\hat{x}_a - \hat{x}_b$.) Then we always have $x_a, x_b \in T$, and setting $\hat{x}_a - \hat{x}_b =: \eta$, this implies,

$$\sup_{\|\eta\|=\hat{\rho}} \|B\eta\| \leq h_T.$$

Using (13.12) with $\rho = \rho_{\hat{T}}$, the assertion follows. ∎

For shape-regular triangulations, we thus have[5]

$$\|B\| \left\|B^{-1}\right\| = \frac{h_T}{\rho_{\hat{T}}} \frac{h_{\hat{T}}}{\rho_T} \leq \kappa \frac{h_{\hat{T}}}{\rho_{\hat{T}}} \leq C_B < \infty. \tag{13.13}$$

13.6 Interpolation error estimate

We now give an important interpolation result that lies at the root of our error analysis. Basically only this one result, applied to the reference triangle, is needed. The rest is done by the affine mapping introduced in the previous section, and the bounds on the Jacobian of that mapping (cf. Theorem 13.2). The following theorem applies to quite general domains. However, we are interested only in analyzing the interpolation on a triangle. Therefore, we formulate the theorem specifically for a generic triangle and piecewise linear interpolation:

Theorem 13.3 *Let T be a triangle, and let $v \in H^2(T)$. Let $\mathcal{I}_T$ be the interpolation operator defined in* (13.4)*. Then there is a constant $C > 0$, such that*

$$\|v - \mathcal{I}_T v\|_{m,T} \leq C|v|_{2,T}, \qquad m = 0, 1, 2. \tag{13.14}$$

Proof (Sketch) We show the inequality for $m = 2$. The corresponding inequalities for $m = 0, 1$ follow trivially from $\|v\|_t \leq \|v\|_2$ for $t = 0, 1$.

Let z_i, $i = 1, 2, 3$ be the vertices of the triangle T that are used for the interpolation. It can be shown that

$$|||v||| := |v|_{2,T} + \sum_{i=1}^{3} |v(z_i)|$$

is a proper norm. Furthermore, one can show that this norm is equivalent to the H^2-norm. This means there exist positive constants C_1, C_2, such that

$$C_1 \|v\|_{2,T} \leq |||v||| \leq C_2 \|v\|_{2,T}. \tag{13.15}$$

Roughly speaking, one can argue that the nodal values at the z_i uniquely specify the linear part of v, and thus we have sufficient information up to the first derivatives, such that, when adding information about the second derivatives via the semi-norm $|\cdot|_{2,T}$, the norm $|||\cdot|||$ is equivalent to the regular $\|\cdot\|_{2,T}$ norm. The first inequality in (13.15) implies that there is a constant $C > 0$, such that

$$\begin{aligned}
\|v - \mathcal{I}_T v\|_{2,T} &\leq C \left(|v - \mathcal{I}_T v|_{2,T} + \sum_{i=1}^{3} \underbrace{|(v - \mathcal{I}_T v)(z_i)|}_{=0} \right) \\
&= C \left(\sum_{|\alpha|=2} \|D^\alpha (v - \mathcal{I}_T v)\|_{0,T}^2 \right)^{\frac{1}{2}} \\
&= C \left(\sum_{|\alpha|=2} \|D^\alpha v\|_{0,T}^2 \right)^{\frac{1}{2}} \\
&= C|v|_{2,T}
\end{aligned}$$

We have used that $(v - \mathcal{I}_T v)(z_i) = 0$ (note that z_i are the nodes used in the interpolation!), as well as the definition of the semi-norm $|\cdot|_{2,T}$. Note that $D^\alpha(\mathcal{I}_T v) = 0$, as $\mathcal{I}_T v \in \mathcal{P}^1$ (i.e. the interpolant is a polynomial of degree one.) ■

[5] Note that $\rho_{\hat{T}}$ and $h_{\hat{T}}$ depend only on the choice of reference element and can be considered fixed constants in this analysis.

We now use this result specifically for the reference element introduced in 13.4, and then get a corresponding interpolation estimate for each element in physical space by appropriate integral transformations (i.e. eqns. (13.7), (13.8)).

First, we define a mesh dependent norm as

$$\|v\|_{m,\mathcal{T}_h}^2 := \sum_{T\in\mathcal{T}_h} \|v\|_{m,T}^2 \,.$$

For functions $v \in H^m(\Omega)$ with $m > 1$, and interpolants $\mathcal{I}_h v \in S_h \subset H^1(\Omega)$ this coincides for $m = 1$ with the usual definition of the norm $\|\cdot\|_{m,\Omega}$. However, for $m > 1$ the interpolant is not in $H^m(\Omega)$. Thus, a "broken" norm is introduced, which sums up the local norms over all elements.

Theorem 13.4 *Let $u \in H^2(\Omega)$, and let $\mathcal{T}_h$ be a shape-regular triangulation. Let the interpolation operator $\mathcal{I}_h u$ be defined such that the restriction to each element $T \in \mathcal{T}_h$ is given by* (13.4)*. Then there holds*

$$\|u - \mathcal{I}_h u\|_{m,\mathcal{T}_h} \le ch^{2-m}|u|_{2,\Omega}, \qquad m = 0, 1, 2,$$

where h is defined as in (13.10)*, and c is a positive constant.*

Proof For the semi-norm $|u - \mathcal{I}_h u|_{m,\mathcal{T}_h}$, we have the following estimate:

$$|u - \mathcal{I}_h u|_{m,\mathcal{T}_h}^2 = \sum_{T\in\mathcal{T}_h} |u - \mathcal{I}_h u|_{m,T}^2 \quad \le \quad C \sum_{T\in\mathcal{T}_h} |\det B_T| \left\|B_T^{-1}\right\|^{2m} |\hat{u} - \mathcal{I}_{\hat{T}}\hat{u}|_{m,\hat{T}}^2$$

where (13.7) has been used. (Here and in the following we often add the subscript T to the Jacobian of the transformation to associate it with a triangle T.)

Note that in the following we always denote by C a generic constant, which is not necessarily the same at each occurence. Using Theorem 13.3, via

$$|\hat{u} - \mathcal{I}_{\hat{T}}\hat{u}|_{m,\hat{T}}^2 \le \left\|\hat{u} - \mathcal{I}_{\hat{T}}\hat{u}\right\|_{m,\hat{T}} \le C|\hat{u}|_{2,\hat{T}} \tag{13.16}$$

(the first inequality is trivial), and the estimate (13.8), we have

$$\begin{aligned}
|u - \mathcal{I}_h u|_{m,\mathcal{T}_h}^2 &\le C \sum_{T\in\mathcal{T}_h} |\det B_T| \left\|B_T^{-1}\right\|^{2m} |\hat{u}|_{2,\hat{T}}^2 \\
&\le C \sum_{T\in\mathcal{T}_h} |\det B_T| \left\|B_T^{-1}\right\|^{2m} |\det B_T|^{-1} \|B_T\|^4 |u|_{2,T}^2 \\
&= C \sum_{T\in\mathcal{T}_h} \left\|B_T^{-1}\right\|^{2m} \|B_T\|^4 |u|_{2,T}^2
\end{aligned}$$

Now insert the identity $1 = \|B_T\|^{2m} \|B_T\|^{-2m}$ into the right-hand side to obtain

$$\begin{aligned}
|u - \mathcal{I}_h u|_{m,\mathcal{T}_h}^2 &\le C \sum_{T\in\mathcal{T}_h} (\|B_T\|^{2m} \|B_T\|^{-2m}) \left\|B_T^{-1}\right\|^{2m} \|B_T\|^4 |u|_{2,T}^2 \\
&= C \sum_{T\in\mathcal{T}_h} (\|B_T\| \left\|B_T^{-1}\right\|)^{2m} \|B_T\|^{4-2m} |u|_{2,T}^2 \\
&\le C \sum_{T\in\mathcal{T}_h} h_T^{4-2m} |u|_{2,T}^2
\end{aligned}$$

where the estimate (13.13) was used. Note that $\|B\| \le h_T \rho_{\hat{T}}^{-1}$ (by Theorem 13.2) has also been used. The constant $\rho_{\hat{T}}$ depends only on the choice of reference element. This finally yields,

$$|u - \mathcal{I}_h u|_{m,\mathcal{T}_h}^2 \le Ch^{4-2m} \sum_{T\in\mathcal{T}_h} |u|_{2,T}^2 = \widetilde{C}h^{4-2m}|u|_{2,\Omega}^2$$

Now we can estimate,

$$\begin{aligned}
\|u - \mathcal{I}_h u\|_{m,\mathcal{T}_h} &= \left(\sum_{s=0}^{m} |u - \mathcal{I}_h u|_{s,\mathcal{T}_h}^2\right)^{\frac{1}{2}} \\
&\leq C\left(\sum_{s=0}^{m} h^{4-2s}|u|_{2,\Omega}^2\right)^{\frac{1}{2}} \\
&\leq Ch^{2-m}|u|_{2,\Omega}
\end{aligned}$$

where we have only kept the leading order error term. ■

Note that the preceding theorem shows in particular

$$\|u - \mathcal{I}_h u\|_{1,\Omega} \leq ch|u|_{2,\Omega}$$

As anticipated in eq. (13.2), we can connect this estimate to the optimality result derived in section 12.2. We thus obtain the main result for the error of the Ritz-Galerkin approximation as

$$\|u - u_h\|_{1,\Omega} \leq ch|u|_{2,\Omega}.$$

Also note that, while we have an interpolation error of $\mathcal{O}(h^2)$ for the L_2-norm (i.e. for $m = 0$), we cannot trivially infer an error of this order for the Ritz-Galerkin approximation, because we have nothing to which we can connect this result. The optimality result for the H^1 norm follows from the fact the we could show boundedness and coercivity in that norm. We have no corresponding result for the L_2 norm. Nevertheless, we would hope that we gain an order of magnitude in convergence when measured in L_2 norm, just like for the interpolation error. This is the topic of the next section.